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Set 15

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Dataset: Monthly Product Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data

Solution :

Code:

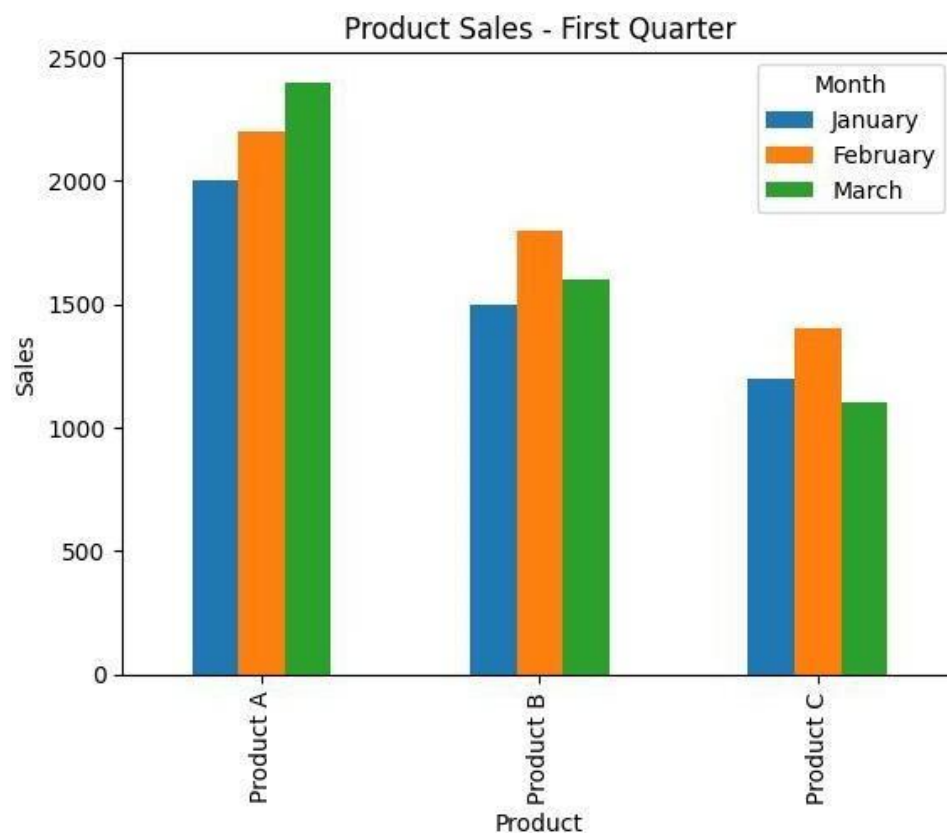
```
Import pandas as pd
```

```
Import matplotlib.pyplot as plt
```

```
Df = pd.DataFrame({  
    'Product': ['Product A', 'Product B', 'Product C'],  
    'January': [2000, 1500, 1200],  
    'February': [2200, 1800, 1400],  
    'March': [2400, 1600, 1100]  
})  
  
Df.set_index('Product').plot(kind='bar')  
  
Plt.xlabel('Product')  
  
Plt.ylabel('Sales')
```

```
Plt.title('Product Sales – First Quarter')
Plt.legend(title='Month')
Plt.show()
Df.set_index('Product').T.plot(kind='area', stacked=True)
Plt.xlabel('Month')
Plt.ylabel('Sales')
Plt.title('Overall Sales Trend')
Plt.show()
Display(df)
```

Output:



Overall Sales Trend

